Función de transferencia

$$F_a(t) = F_z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$F_z(t) = \frac{P_a(t) - P_p(t)}{Z}$$

$$F_L(t) = \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt$$

$$F_C(t) = C\,\frac{dP_p(t)}{dt}$$

$$F_R(t) = \frac{P_p(t)}{R}$$

Transformada de Laplace

$$F_a(s) = F_z(s) + F_L(s) = F_C(s) + F_R(s)$$

$$F_z(s) = \frac{P_a(s) - P_p(s)}{Z}$$

$$F_L(s) = \frac{P_a(s) - P_p(s)}{Ls}$$

$$F_C(s) = Cs\,P_p(s)$$

$$F_R(s) = \frac{P_p(s)}{R}$$

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Procedimiento algebraico

$$F_Z(s) + F_L(s) = F_C(s) + F_R(s)$$

$$\frac{P_a(s) - P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{Ls} = Cs P_p(s) + \frac{P_p(s)}{R}$$

$$\frac{P_a(s)}{Z} - \frac{P_p(s)}{Z} + \frac{P_a(s)}{Ls} - \frac{P_p(s)}{Ls} = Cs P_p(s) + \frac{P_p(s)}{R}$$

$$P_a(s)\left(\frac{1}{Z} + \frac{1}{Ls}\right) = P_p(s)\left(Cs + \frac{1}{R} + \frac{1}{Ls} + \frac{1}{Z}\right)$$

$$P_a(s)\frac{Ls+Z}{LZs} = P_p(s)\frac{CLRZs^2 + LZs + Rz + RLs}{RLZs}$$

$$P_a(s)\frac{Ls+Z}{LZs} = P_p(s)\frac{CLRZs^2 + L(Z+R)s + Rz}{RLZs}$$

$$\frac{P_p(s)}{P_a(s)} = \frac{\frac{Ls+Z}{LZs}}{\frac{CLRZs^2 + L(Z+R)s + RZ}{RLZs}}$$

Resultado

$$\frac{P_p(s)}{P_a(s)} = \frac{LRs + Rz}{CLRZs^2 + L(Z+R)s + Rz}$$

Estabilidad del sistema en lazo abierto

Hipotenso $\quad CLRZs^2 + L(Z+R)s + Rz = 0$

$Z = 0.001$
$C = 3.5$
$R = 1$
$L = 0.001$

$a = CLRZ$
$b = L(Z+R)$
$C = RZ$

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$\lambda_1 = -14.428 + 8.805j$ respuesta estable y subamortiguada
$\lambda_2 = -14.428 - 8.805j$ a un escalón unitario de entrada

normotenso

$z = 0.01$
$C = 1.5$
$R = 1.920$
$L = 0.001$

$\lambda_1 = -12.061$
$\lambda_2 = -55.271$

respuesta estable y sobreamortiguada a un escalon unitario de entrada

hipertenso

$z = 0.01$
$C = 1.000$
$R = 1.000$
$L = 0.0011$

$\lambda_1 = -11.126 + 1.6843i$
$\lambda_2 = -89.893 - 1.6843i$

respuesta estable y sobreamortiguada a un escalon unitario de entrada

Error en estado estacionario

$$e(s) = \lim_{s \to 0} s P_a(s) \left(1 - \frac{P_p(s)}{P_a(s)}\right) = e(s)$$

$$\lim_{s \to 0} s \frac{1}{s} \left(1 - \frac{LRs + Rz}{CLRz s^2 + L(z+R)s + Rz}\right) = 0$$

$$e(t) = 0 V$$

Modelo de ecuaciones integro-diferenciales

$$F_z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$\frac{P_a(t)}{z} + \frac{1}{L}\int [P_a(t) - P_p(t)]dt - C\frac{dP_p(t)}{dt} = \frac{P_p(t)}{R} + \frac{P_p(t)}{z}$$

$$\left(\frac{1}{R} + \frac{1}{z}\right) = P_p(t) = \frac{P_a(t)}{z} + \frac{1}{L}\int [P_a(t) - P_p(t)]dt - C\frac{dP_p(t)}{dt}$$

$$P_p(t) = \left[\frac{P_a(t)}{z} + \frac{1}{L}\int [P_a(t) - P_p(t)]dt - C\frac{dP_p(t)}{dt}\right]\frac{Rz}{z+R}$$

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